

Getting the shade right first time at batch change: our proposal

To: Special Projects Lead, Procurement · **Cc:** Head of IT & OT Procurement · **Date:** 3 October 2026

Thank you for the call. Here is what we propose, in plain terms.

What we heard. When an SKU is run again, the first fired tile often doesn't match the master. Your team then edits the print file by eye, fires 6–7 variants, checks them in the lab, and repeats. Each round takes 6–7 hours, and the line produces thousands of square metres in the wrong shade in the meantime.

What we propose. Your lab already measures the colour difference precisely. What's missing is a reliable answer to "how much do I change the file?" for *this* batch. We will change the way the 6–7 test tiles in a round are chosen, so that **one round tells us exactly how this batch responds to each ink**. A small program then calculates the file change for the whole design at once. Your QC team still approves every change on the measured result. There is **no new hardware**, and **nothing connects to your printers**. The program runs on a PC in your lab and uses the instrument you already have.

Two things to be clear about up front. First, the *recommendation* takes minutes, but one test firing will always be needed: the colour only appears in the kiln. Our goal is to bring most batch changes down from several rounds to one or two. If a new batch's material can be test-fired before the line switches over, the correction can be ready before the switch, and almost no wrong-shade tiles get made at all. We will check this with you. Second, in week 2 we check whether your kiln gives the same shade twice from the same file. If it doesn't, no file change can fix the problem, and we will tell you so and stop.

What we need from you.

1. A second call **with your plant QC head and a design-team lead**: they are the people who will use this.
2. The master tile and a few variation tiles shipped to us in week 1, plus original print files for one product.
3. Access to the UAE plant for a visit, one named QC contact, and two kiln slots for test tiles in the first three weeks.
4. Any existing records of past correction rounds, and what an off-shade square metre costs you (downgraded, scrapped or held).

Cost.

Stage	What you get	Price
Stage 1: Proof on your tiles (3 weeks)	Site visit, kiln repeatability check, first live correction on one product, measured by your own lab	\$25,000 fixed (half credited to stage 2 if you continue)
Stage 2: Pilot (9 weeks)	Working tool in one plant, tested on 20 real batch changes against your current method	\$110,000 fixed: 30% at start, 30% at week 7, 40% only if the agreed result is met
Rollout (optional)	Each additional line	\$12,000 setup + \$3,000 per line per month

What it's worth. On our working assumptions (3 batch changes a month per line, about 3,500–5,500 m² less off-shade output each time, at a 30% loss on a \$6/m² tile), the saving is roughly **\$230,000–\$360,000 per line per year**. We will check these assumptions with your numbers in stage 1 before you commit to stage 2.

When you'll see a result.

- **Week 2:** whether your kiln is consistent enough for this to work (a clear go or stop).
- **Week 3:** a real product corrected in one round, confirmed by your lab.
- **Week 7:** results on 10 products.
- **Week 12:** the go or no-go decision on rollout, with measured numbers.

We would like to schedule the second call with your QC and design leads next week.